

Prakhar Agrawal

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Education

Bachelor of Technology in Computer and Communications Engineering

JK Lakshmipat University, Jaipur, Rajasthan

2023-2027

CGPA: 7.88

Senior Secondary (12th), CBSE

Maheshwari Public School, Pratap Nagar, Jaipur, Rajasthan

2022

79.2%

Secondary (10th), CBSE

Maheshwari Public School, Pratap Nagar, Jaipur, Rajasthan

2020

91.8%

Experience

Machine Learning Intern

May-June 2025

LUSIP-2025, LNMIIT Jaipur

Worked on the hardware implementation of a Yoga Pose Recognition system using Deep Learning to understand system architecture, evaluate suitable ML/DL models, and find low-power solutions refined for Raspberry Pi 4 deployment. Applied these insights to develop a real-time American Sign Language Recognition system using a Random Forest model for efficient Raspberry Pi-based edge deployment.

Projects

CodeCampus – Coding Platform for Colleges

[GitHub Link](#)

Tech: React.js, Tailwind CSS, Node.js, Express.js, MongoDB, JWT, Bcrypt, NodeCron

Built a full-stack MERN-based coding platform and LMS for educational institutions featuring real-time code execution, contest/lab management, leaderboards, gamification, and TA/instructor dashboards for performance tracking and academic workflow automation.

American Sign Language Recognition using Machine Learning on Raspberry Pi

[GitHub Link](#)

Tech: Python, MediaPipe, OpenCV, Machine Learning (Random Forest, scikit-learn), Raspberry Pi

Developed a real-time American Sign Language (ASL) recognition system using MediaPipe-based hand landmark extraction and Random Forest classification, achieving 94%+ accuracy on a custom 9,000+ image dataset; optimized for Raspberry Pi deployment as a portable, low-cost assistive solution.

Skills

- **Programming Languages:** C, Python, VHDL
- **Web Development:** React.js, Tailwind CSS, Node.js, Express.js, MongoDB
- **Embedded Systems & FPGA:** Basic Circuit Design, Arduino
- **Tools:** Git, Xilinx Vivado

Certifications

- C Programming for Everybody | University of Michigan | Coursera
- Red Hat System Administration I (RH124 - RHA) | Red Hat Academy
- Hands-on Training, Research Aspects and Demonstrations in 5G Use Case Lab | AICTE Training and Learning (ATAL) Academy | Punjab Engineering College